

### Proof by Contrapositive

Proof by contrapositive is just *a special case of* proof by contradiction in the sense explained below:

Let  $H$  be the hypothesis (may be composed of a number of statements) and  $C$  be the intended conclusion. Then, in this technique,  $\sim C$ , the negation of  $C$ , is assumed to be true as is done in Proof by contradiction. *The proof of  $C$  from  $H$  follows if we are able to derive  $\sim H$ , and hence, a special contradiction*, viz.  $H$  and  $\sim H$  both become true. In the Proof by contradiction', the contradiction may be constituted of  $S$  and  $\sim S$ , where  $S$  may be *any* statement.

**Example:** For every  $n \in \mathbb{Z}$ , if  $n^2$  is even, then  $n$  is even.

We prove the result *by contrapositive* method. Now  $H$ : ( $n \in \mathbb{Z}$ , and  $n^2$  is even),  $C$ :  $n$  is even. Let us assume  $\sim C$  as true, i. e.  $n$  is odd, which means  $n = 2k + 1$  for some  $k \in \mathbb{Z}$ . Then  $n^2 = (2k+1)^2 = 4k^2 + 4k + 1$ . Hence  $n^2$  is odd. Thus, we get  $\sim H$ . Initially,  $H$  is assumed to be true. Thus, we arrive at the contradiction that both  $H$  and  $\sim H$ , being simultaneously true. Hence, the proof by contrapositive follows.

### III) Proof by Cases

The technique is explained through the following example

**Example:** Show there are no solutions for integers  $x$  &  $y$  such that  $x^2 + 3y^2 = 8$

Consider the following cases

Case (i)  $|x| \geq 3$  then  $x^2 > 8$  Hence,  $x^2 + 3y^2 > 8$ , and not  $= 8$

Case (ii)  $|y| \geq 2$ , then  $3y^2 > 8$  Hence,  $x^2 + 3y^2 > 8$ , and not  $= 8$

Thus  $|x| \leq 2$  and  $|y| \leq 1$ .

The only possibilities for integers  $x$  and  $y$  satisfying these two inequalities are

Case (iii)  $x = -2, -1, 0, 1, \text{ or } 2$  &  $y = -1, 0, 1$ . Thus

possible values for  $x^2 = 0, 1, 4$

possible values for  $3y^2 = 0, 3$

Then Maximum possible value of  $(x^2 + 3y^2) = 7$ , and not 8 as required.

### IV) Existence Proofs

#### a) Constructive

$(\exists X) P(x)$  is shown by the existence of  $b$  such that  $P(b)$  is true

Problem: show that any positive integer can be written as sum of two cubes in two different ways

$$\text{Solution: } 1729 = 10^3 + 9^3 = 12^3 + 1^3$$

### b) Non-constructive Existence Proof

Problem: For some irrationals  $x$  and  $y$ , the number  $x^y$  is rational

Solution: We know  $(\sqrt{2})$  is irrational. Then either (i)  $(\sqrt{2})^{\sqrt{2}}$  is rational, or (ii)  $(\sqrt{2})^{\sqrt{2}}$  is irrational. In case of (i) the problem is solved, because for  $x=y=\sqrt{2}$ , irrational, we get  $(\sqrt{2})^{\sqrt{2}}$  is as rational. In case (ii) if  $(\sqrt{2})^{\sqrt{2}}$  is irrational then  $[(\sqrt{2})^{\sqrt{2}}]^{\sqrt{2}} = 2$  is rational, for  $x=(\sqrt{2})^{\sqrt{2}}$  and  $y = \sqrt{2}$ , both irrationals. (In the non-constructive existence, the actual values of irrational numbers are not calculated)

## V) Proof by Mathematical Induction

A proof by mathematical induction is used to prove *infinitely many* statements, if

- i) all of them happen to be True, and
- ii) which can be indexed by elements of  $\mathbb{N}$ , the set of Natural numbers.

For example, the statement  $S_n$ , **the sum of first  $n$  Natural numbers =  $n(n+1)/2$** , is true for all  $n \in \mathbb{N}$ .

Then, the set  $\{S_n : n \in \mathbb{N}\}$  is an infinite set of statements indexed by members of  $\mathbb{N}$ , and also each  $S_n$  is true. *The truth of all  $S_n$ 's can be established by using Principle of Mathematical Induction to be explained below.*

The method of Mathematical Induction has following three major steps:

- i) Initial/ base step
- ii) Induction Hypothesis step
- iii) Inductive step

Some people combine (ii) Induction Hypothesis step and (iii) Inductive step into a single step and call it *Inductive Step*. **We explain the method through applying it to prove truth of infinitely many statements  $\{S_n : n \in \mathbb{N}\}$ , where the  $n^{\text{th}}$  statement is  $S_n = \text{the sum of first } n \text{ Natural numbers} = n(n+1)/2$ .**

We may note that here we are *not* dealing with a single statement, but *infinitely many* statements:

Prove that

$$S_1 = \text{sum of first one natural number} = 1(1+1)/2$$

$$S_2 = \text{sum of two natural numbers} = 2(2+1)/2$$

. . . . .  
. . . . .

$S_{10}$  = sum of first ten natural numbers =  $10(10+1)/2$

.....  
.....

$S_{1000}$  = sum of first one thousand natural numbers =  $1000(1000+1)/2$

.....

*But the proof is not required for each of the above separately. The proof is for general  $n$ , and hence for all of them simultaneously.*

**Base/ initial step:** show for  $n=1$ , i. e.  $S_1$  is true, i. e. sum of first one natural numbers =  $1(1+1)/2 = 1$

But this is true. Hence, for this problem, base is established/ shown as true.

**Remarks:** some time instead of  $\{S_n: n \in \mathbb{N}\}$ , the set of statements to be proved is of the form  $\{S_n: n \in \mathbb{N} \text{ and for all } n \geq t > 1, \text{ for some } t \in \mathbb{N}\}$ . Then, base step starts with  $S_t$ .

**Induction Hypothesis:** Let, for any  $k > 1$ ,  $S_k$ : sum of first  $k$  natural numbers =  $k(k+1)/2$  be assumed to be true.

**Inductive Step:** With the help of base step result and the assumption of Induction hypothesis, we attempt to show that  $S_{k+1}$ : sum of first  $(k+1)$  natural numbers =  $(k+1)(k+2)/2$ .

*If we are able to show it, the Principle of Mathematical Induction states that*

$S_n$ : sum of first  $n$  natural numbers =  $n(n+1)/2$  is true  $\forall n \in \mathbb{N}$ .

So, consider

$$S_{k+1} = 1 + 2 + \dots + k + k+1 = (1 + 2 + \dots + k) + (k+1) = S_k + (k+1)$$

(By induction hypothesis) =  $k(k+1)/2 + (k+1) = (k+1)(k/2 + 1) = (k+1)(k+2)/2$ .

Hence, by Principle of Mathematical Induction

$S_n$ : sum of first  $n$  natural numbers =  $n(n+1)/2$  is true  $\forall n \in \mathbb{N}$

## VI) Proof of Equivalences

An equivalence in a statement to be proved is indicated by the phrase 'if and only if', which is also written as 'iff'.

**Example:** For  $x, y \in \mathbb{N}$ , the set of natural numbers, then  $x > y$  iff  $x^2 > y^2$

Ans. The proof consists of two parts:

- i) By assuming  $x, y \in \mathbb{N}$ , and  $x > y$ , to show that  $x^2 > y^2$ , and
- ii) By assuming  $x, y \in \mathbb{N}$ , and  $x^2 > y^2$  to show that  $x > y$

**For part (i) (this part is generally called ‘if part’:** let  $x > y$ , as  $x \in \mathbb{N}$ , therefore,  $x > 0$ , multiplying both sides by  $x$ , we get  $x^2 = x \cdot x > x \cdot y$

Similarly, as  $y > 0$ , from  $x > y$ , we get  $y \cdot x > y \cdot y = y^2$

Using commutativity, and associativity of ‘=’ etc. of ‘.’, we get  $x^2 > y^2$

**For part (ii) (this part is generally called ‘only if part’:** let  $x, y \in \mathbb{N}$ , and  $x^2 > y^2$ , to show that  $x > y$ .

The argument is similar, except that, initially, we will state that as each of  $x > 0$  and  $y > 0$ , therefore each of  $(1/x) > 0$  and  $(1/y) > 0$ ,

Then by replacing  $x$  by  $(1/x)$  and  $y$  by  $(1/y)$  throughout the argument, ultimately, we get  $x > y$ .

**Remarks:** Sometimes, the two parts ‘if part’ and ‘only if part’ may be combined. *For example, to show that the integer  $n$  is even if and only if  $2n+3$  is odd*

The proof may be written as follows, in which each of the symbols (i) ‘ $\Leftrightarrow$ ’ and (ii) ‘*iff*’ is used for phrase: ‘if and only if’

**Proof:**  $n \text{ is even} \Leftrightarrow n=2k \text{ for some } k \in \mathbb{Z} \Leftrightarrow 2n=2(2k) \text{ for } k \in \mathbb{Z} \Leftrightarrow 2n+3=4k+3 \Leftrightarrow 2n+3=4k+2+1 \Leftrightarrow 2n+3=2(2k+1)+1, \Leftrightarrow 2n+3=2K'+1 \text{ for } K'=2k+1 \Leftrightarrow 2n+3 \text{ is odd} \Leftrightarrow (2K'+1) \text{ is odd for } K'=2k+1. \text{ But R.H.S is true. Hence the proof is complete.}$

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## UNIT 3 RELATIONS AND FUNCTIONS

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### Structure

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### 3.0 OBJECTIVES

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After going through this unit, you should be able to:

- distinguish between concepts of relation and function;
- use the notation for relation;
- explain different properties like reflexivity, symmetry and transitivity of relations;
- apply the concepts of equivalence relation, equivalence class, and partition of a set;
- identify partial order relation, and partially ordered set;
- draw Hasse diagram for a partially ordered set;
- apply operations, like composition, inverse and union, to relations;
- apply operations on functions; and
- determine various types of function: injective, surjective and bijective.

### 3.1 INTRODUCTION

In Unit 1, we have already talked of *set-relations* (i. e., *relations on sets*)—like  $\subseteq$  (subset),  $\subset$  (proper subset), and  $=$  (equality) of sets—between pairs of sets; and *number-relations*—like  $<$  (less than),  $>$  (greater than), and  $=$  (equal to)—between pair of numbers. In everyday life, we talk of relations like of ‘father to daughter’, ‘mother to son’, ‘sister to brother’, ‘teacher to student’, and ‘employer to employee’. In economics, occurrence of the concept of *relation* is almost universal, e. g. (item, cost-of-production), (item, sales-price), (income-of-a-family, expenditure) to name a few.

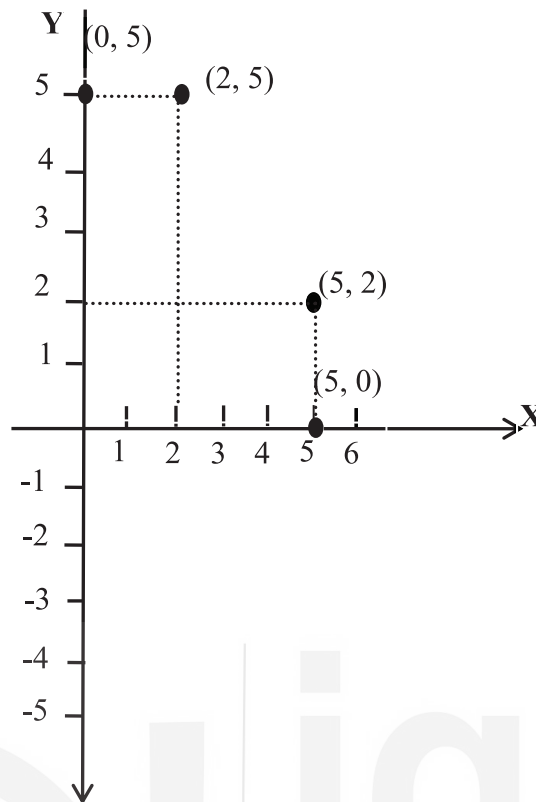
In most of such cases, *there is a sense of direction/order from the first to the second* in the sense that if X is mother of Y, then Y is not mother of X, i.e., X and Y cannot be exchanged in the relation of *is-mother-of*. Similarly, for the relation of *is-less-than* ( $<$ ), we have  $4 < 5$ , and we cannot exchange roles of 4 and 5 for the relation ‘ $<$ ’ to be true. Except in *special cases* of relations like *equal-to* ( $=$ ) in numbers, or the relation of *sibling* (persons, male or female, having same pair of parents) among human beings, the *order* of arguments involved in a relation, in general, has to be taken into consideration.

Also, a relation need *not* be between two *similar* sets, as discussed above: between two sets of sets, between two sets of numbers, or between two sets of human beings. It may be between two different types of sets. For example, the relation of *age-of-person* is *from* a set of human beings *to* a set of numbers. This leads us to the concepts ‘*ordered set*’ and ‘*cross-product of two sets*’, which are discussed next.

a) **Ordered Set:** Earlier, in Unit 1, it was pointed out that in a set, order of occurrence of elements is irrelevant:

For example,  $\{1, 2, 3, 4\} = \{4, 2, 1, 3\} = \{3, 2, 4, 1\} = \dots$

However, there are situations, as mentioned just above, in which the change of order of occurrence of the elements, changes the situation. For example, in co-ordinate geometry (Fig. 3.1), the location of the point (2, 5) is different from the location of the point (5,2).



**Fig. 3.1: Location of Points in Co-ordinate Geometry**

Such a set, in which order of occurrence of elements is significant, is called an **ordered set**. In the notation of ordered sets, the elements are enclosed between **parentheses**, instead of **braces/ curly-brackets**. Thus

$$(a, b) \neq (b, a), \text{ but } \{a, b\} = \{b, a\}.$$

b) **Cross-product of two sets X and Y**, for  $X \neq \emptyset \neq Y$ , denoted by  $X \times Y$  is the set of all ordered pairs  $(x, y)$  with  $x$  belonging to  $X$  and  $y$  belonging to  $Y$ , i. e.

$$X \times Y = \{(x, y): x \in X, y \in Y\}.$$

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## 3.2 RELATION

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The concept of 'relation' is one of a few fundamental concepts of mathematics. We begin its discussion with its formal/ mathematical definition and illustrations through some well-known examples.

### 3.2.1 Definition, Notation and Illustrations

In Mathematics, a *relation from a set X to a set Y* is defined as a **subset** of the cross-product  $X \times Y$ . Earlier mentioned relations like *is-mother-of* or *is-less-than* can be discussed as subsets of cross-products of some sets as follows.

**Notation (in Mathematics)**

If *Sheela* is mother of *Shivam*, then the fact may be expressed as

*Sheela is-mother-of Shivam.*



In mathematics, it is preferred to denote the fact as

***(Sheela, Shivam) ∈ is-mother-of,***

where the relation ***is-mother-of*** is considered as ***the set*** of all the pairs (m, c), for which m is mother of c.

In one of methods of mathematical shorthand, *is-mother-of* is denoted in set notation as

***is-mother-of*** = {(m, c): m ∈ X, c ∈ Y, and m is mother of c} ⊆ X × Y,

where X = Y = set of human beings. ... ..(i)

(The above form (i) is called ***set-builder form***, in which the name of the property viz. ***is mother of*** is used)

Then the fact *Savitri is not mother of Mohan*, is denoted as

***(Savitri, Mohan) ∉ is-mother-of.***

Similarly, the relation '*is-less-than*' (<) on N, may be considered as a subset of a cross-product—by taking X = Y = N = {1, 2, 3, ...}—as follows

***is-less-than*** = {(1, 2), (1, 3), (1, 4), ..., (2, 3), (2, 4), (2, 5), ..., (3, 4), (3, 5), (3, 6), ...}, or more systematically as

***is-less-than*** = {(1, 2), (1, 3), (1, 4), (2, 3), (1, 5), (2, 4), (1, 6), (2, 5), (3, 4), (1, 7), (2, 6), (3, 5), (1, 8), (2, 7), (3, 6), (4, 5), (1, 9), (2, 8) ...} ⊆ N × N. ... .. (ii)

Thus, (1, 4) ∈ *is-less-than*, and (4, 1) ∉ *is-less-than*.

(The above form (ii) is called ***roster form***, in which each pair for which property holds is mentioned explicitly instead of the name of the property ***is-less-than***)

The ***set-builder form*** of *is-less-than* is given by

***is-less-than*** = {(x, y): x, y ∈ N, and x < y}

***Argument(s) of a relation:*** the values supplied to a relation are called *arguments* of the relation. The relations discussed just above involve ***exactly two*** arguments: two human beings, two numbers, or two sets. However, some relations may involve ***just one argument***, e. g. *is-male* or *is-female*: for example,

*is-male* (Mohan) is true, and *is-female* (Mohan) is false.

A ***unary relation*** is also called a *property* of its argument, e. g. *is-male* is a property of Mohan, *is-female* is not a property of Mohan; *is-prime-number* is a property of 19, *is-composite-number* is not a property of 19.

Some relations may involve *more than two* arguments, e. g. the relation ***is-integer-strictly-between*** involve ***three*** integers, such that 4 *is-integer-strictly-between* 7 and 2 is *True*. But 4 *is-integer-strictly-between* 7 and 9 is *False*. Also, 4 *is-integer-strictly-between* 1 and 3 is also *False*. We may express the relation, in set-builder form as ***is-integer-strictly-between*** = {(x, y, z): *either* y < x < z, or



$y > x > z$ . This relation involves **three** arguments, and such relations are called **ternary** relations.

Similarly, the relation **are-parents-of** involves three arguments, and may be defined as **are-parents-of** =  $\{(x, y, z): x, y, z \text{ are human beings; and } x \text{ \& } y \text{ are parents of } z\}$ . In economics, the (item, cost-production, sales-price) is a ternary relation.

Similarly, we may have relations involving four, five or more arguments. For example, the relation of **time-table** of a high-school may involve **four arguments** as follows.

**Time-table** =  $\{(class, subject, subject-teacher, room-number): class \in \{VI, VII, VIII, IX, X\},$

$subject \in \{Science, Maths., Hindi, English, \dots\},$

$subject-teacher \in \{Ms. Geeta, Mr. John, Mr. Abdul, \dots\},$

$room-number = \{1, 2, 3, \dots\}\}$ .

This is an example of **quaternary relation**. In general, a relation  $R$  may involve  $n$  sets, say  $X_1, X_2, \dots, X_n$ , then  $R \subseteq X_1 \times X_2 \times \dots \times X_n$ , and  $R$  is called  **$n$ -ary relation**.

However, most of the time, we will be discussing only relations, like *is-mother-of* or *is-less-than*, each of

which requires exactly **two arguments**. A relation which involves **two** arguments is called a **binary** relation.

Unless, we explicitly state otherwise, the term '**relation**' will stand for '**binary relation**'.

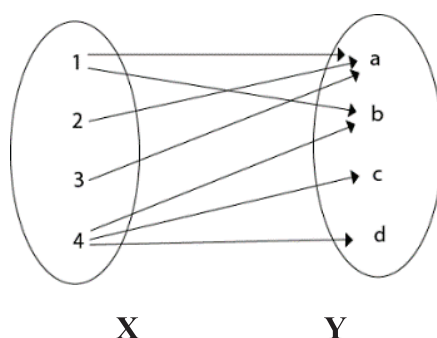
### Some more definitions

- **Domain, Codomain & Range of a relation:** For a **binary relation**  $R$  from  $X$  to  $Y$ , i. e.  $R \subseteq X \times Y$ , the set  $X$  is called the **domain** of  $R$  and the set  $Y$  is called the **codomain** of  $R$ . For example, consider the relation of  $R = (Person, Number)$ , number stands for 10-digit mobile number. Then **domain** of  $R$  is the set of all persons, and the **codomain** of the relation  $R$  is the set of all 10-digit natural numbers. The set of all 10-digit numbers, which are *used* as 10-digit mobile numbers of some persons, constitute the **range** of  $R$ .
- **Image of an element of the domain:** For the relation  $R \subseteq X \times Y$ , for  $x \in X$  (domain), and  $y \in Y$  (codomain), if  $(x, y) \in R$ , then an image of  $x$  under relation  $R$  is  $y$ . For example, If *Sheela* is mother of *Shivam*, then under the relation *is-mother-of*, one of the **images** of *Sheela* is *Shivam*.
- **Relation on a set  $X$ :** For a **binary relation**  $R$ , if  $codomain = domain$  (i. e.,  $R$  is from  $X$  to  $X$ ), then  $R$  is said to be a **relation on  $X$** .

- **Onto relation:** For a **binary relation**  $R$  from  $X$  to  $Y$ , we know  $R \subseteq X \times Y$ . It is not necessary that every element of  $Y$  should occur as second component of some ordered pair in  $R$ . Consider the relation of  $(Person, Number)$ , number stands for 10-digit mobile number. Then, it is not necessary that every 10-digit number is a mobile number of someone. Thus,  $(Person, Number)$  is **not an onto** relation. But the relation **is-mother-of** is **onto**, because, if  $P$  is a person, then  $P$  must have a mother, say  $M$ , Thus,  $P$  occurs as second component of  $(M, P)$ .
- **One-to-one relation:** A **binary relation**  $R$  from  $X$  to  $Y$  is said to be **One-to-one relation** if any two distinct elements of the domain cannot have the same image. One of the examples for **one-to-one relation** is  $(Person, Mobile-Number)$ : no two different persons can have the same mobile number.
- **Empty Relation:** A relation  $R$  from a set  $X$  to a set  $Y$  is called **empty relation**, if no element of  $X$  is related under  $R$  to any element of  $Y$ , i. e.  $R = \emptyset \subset X \times Y$ . For example,  $X = \text{set of decimal digits} = \{0, 1, 2, 3, \dots, 9\}$ , and the relation  $R$  is defined **on**  $X$  as  $R = \{(a, b): a - b = 10\}$ . This is empty relation, as no pair  $(a, b)$  of decimal digits satisfies the condition  $a - b = 10$ .
- **Universal Relation:** A relation  $R$  from a set  $X$  to a set  $Y$  is called **Universal Relation**, if each element of  $X$  is related to every element of  $Y$ , i.e.,  $R = X \times Y$ .
- The empty relation and the universal relation are sometimes called **trivial relations**.

**Notations re-visited:** In general, for  $R$ , a relation on a set  $X$ , the fact that  $x$  is related to  $y$  under  $R$  may be denoted by either of the notations **(i)  $(x, y) \in R$ , or (ii)  $x R y$** . Apart from the above-mentioned **set notations** for relations, two more forms are popular, and will be discussed in details under (i) **Graphs** and (ii) **Matrices**. Here, we just give glimpses of the notations through examples. Fig.3.2 explains how to obtain **graphic representation** of a relation, say,

$R = \{(1, a), (1, b), (2, a), (3, a), (4, b), (4, c), (4, d)\}$  from the set  $X = \{1, 2, 3, 4\}$  to the set  $Y = \{a, b, c, d\}$



**Fig. 3.2: Graphic Representation of a Relation**